CORNERSTONE
PDK Monitor

User Manual

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Overview

The CORNERSTONE PDK Monitor is a single-file, browser-based dashboard for managing and monitoring Process Design Kit (PDK) building blocks (BBs), design documentation, and fabrication data for silicon photonics platforms. No server, no install — open `pdk-monitor.html` in any modern browser.

1. Quick Start

1. Open `pdk-monitor.html` in Chrome, Firefox, or Safari (latest versions).
2. Accept the licence terms on first launch.
3. Click **About** in the header to confirm the version.
4. Use the **platform tabs** (SOI 220 nm, SOI 340 nm, TFLN 300 nm, …) to navigate.
5. Click any BB card to view details, upload files, or add test data.
6. Click **Insight** to open the AI agent (requires an API key — see §9).

All data is stored locally in your browser's `localStorage`. To share or back up, use **Export JSON** or publish to a GitHub repository (see §8).

2. Header Controls

Button	Function
Refresh	Reload model from localStorage
Search box	Full-text search across BBs, platforms, contributors
Status / Category dropdowns	Filter the BB list
Useful Links	Configurable chip links (set in About modal)

Button	Function
GitHub	Configure GitHub PAT and identity
Insight	Open the AI Agent (see §9)
About	Version info, notes, editable links and logo
Export	Export dashboard JSON

Storage indicator (bottom-left of header): shows localStorage usage. Turns amber at 70% and red at 90%.

3. Platform Tabs

Each tab represents one fabrication platform (e.g. SOI 220 nm). Tabs show:

- Platform name and material (Si / SiN / LN)
- BB count and completion statistics
- A progress bar coloured by status

Adding a platform: Click **+ Platform** (Admin only).

Reordering: Drag tabs left or right.

Platform settings: Click the pencil icon on any tab to edit name, material, waveguide thickness, and default waveguide width.

4. Building Block (BB) Cards

Each BB is shown as a card with:

- **Name** and **type icon** (waveguide, MMI, ring resonator, grating coupler, etc.)
- **Status dot**: Available (green) · In progress (amber) · Planned (blue) · Discontinued (red)
- **Material badge**: Si · SiN · LN
- **EDA tools** supported (IPKISS, gdsfactory, KLayout, etc.)
- **File count** (GDS, S-matrix, scripts, images)
- **Contributor** chip
- **CDI axes** (Chip / Wafer / Assembly maturity, 1–4)

Click a card to open the **BB Detail** panel on the right side.

5. BB Detail View

The detail panel has five tabs:

Details

- Edit name, description, type, category, status, contributor
- Set CDI maturity levels (chip, wafer, assembly)

- Tag EDA tools where a PDK cell exists
- Assign to a platform variant

Files

Upload or link:

- **GDS / OASIS** layout files
- **S-matrix** data (Touchstone `.s2p`, `.s4p`, CSV)
- **Images** (SEM, cross-section, layout renders)
- **Scripts** (Python, KLayout, gdsfactory cells)
- **Documents** (PDF datasheets, README)

Each file can be opened, downloaded, or deleted. GDS files render a layer-coloured preview inline.

Tests

- Add test entries with date, contributor, and pass/fail status
- Attach raw data files to each test
- View per-test metrics (insertion loss, return loss, bandwidth, etc.)

AI

AI-generated summary of this BB (requires API key — see §9.2). Shows:

- Executive sentence (40 words max)
- Deep-dive analysis (1–3 paragraphs, citing only measured values)
- Cached after first generation; click **Refresh** to regenerate

Parameters

Structured parameter table (waveguide width, gap, coupling length, etc.) editable per sub-entity (e.g. per waveguide width or platform variant).

6. Data & File Management

Uploading files

Drag files onto a BB card or use the upload button in the Files tab. Supported formats: GDS, OASIS, s2p, s4p, CSV, TSV, PNG, JPEG, PDF, PY, KPY, LYML, and more.

GDS viewer

GDS files render inline with layer colours from the configured GDS layer map. Hover a shape for layer name and description. The layer map is editable in **Manage Lists > GDS Layers**.

S-matrix viewer

Upload a Touchstone file (`.s2p`, `.s4p`, `.sNp`) or CSV containing S-parameter sweeps. The viewer plots |S| vs wavelength/frequency for each port pair.

Python script runner

Scripts tagged as Python can be run in-browser via a Pyodide sandbox. Output is shown in a console panel. For scripts that need native binaries or large datasets, use **Run locally** to download a self-contained folder.

7. Multi-User Pool

The pool system lets multiple contributors maintain their own slices of the dashboard in separate GitHub repository folders (`users/<id>/`), which are merged on fetch.

Button	What it does
Fetch all sources	Fetches the main repo + all enabled user slices and merges
Fetch user repos	One-click fetch of all registered user repositories
Discover user slices	Scans the `users/` folder of the configured repo for new contributors
Fetch my slice	Fetches only your own `users/<id>/` folder

User-admin controls (for contributors managing their own slice):

- Add / remove BBs in your slice
- Orphan pruning on publish (removes BBs from the repo that are no longer in the local model)
- **Wipe my repo folder** — removes your entire `users/<id>/` from GitHub

Duplicate BBs (same name across platforms or slices) are flagged in the **Duplicate Resolver**, where you can choose which version to keep or merge them as variants.

8. GitHub Integration

Setting up your GitHub token

7. Click the GitHub button in the header.
8. Paste a GitHub Personal Access Token (PAT) with `repo` scope (fine-grained: Contents read/write).
9. Optionally enter your GitHub username / email and display name.
10. Tick **Remember token** to persist in `localStorage` (stays on this device only, sent only to `api.github.com`).

Publishing

Use **Publish** (Admin) to commit the current `dashboard.json` and all file changes to the configured repository. The dashboard uses the real-files architecture — each uploaded binary lives as its own file; the JSON only stores metadata references.

9. AI Insight Agent

The AI Insight agent is powered by any Claude, OpenAI, or OpenAI-compatible API endpoint (including Ollama and LM Studio for local models).

9.1 Configuration

Click **Insight > Configure** or open Insight and choose **Settings**.

Field	Notes
API endpoint	Default: `https://api.anthropic.com/v1/messages`. For OpenAI use `https://api.openai.com/v1/chat/completions`. For Ollama: `http://localhost:11434/v1/chat/completions`
Model	`claude-sonnet-4-5`, `gpt-4o`, `llama3`, etc.
API key	Your `sk-ant-...` or `sk-...` key. Stored only in this browser's localStorage
Monthly token cap	Hard limit on input+output tokens per calendar month
Egress mode	metrics-only (safest) · metrics + extracts (4 KB/file) · full (50 KB/file)
Prompt style	Brief · Balanced · Deep (process-tolerance interpretation)

Click **Test connection** to verify the endpoint, model and key before saving.

9.2 BB Summariser

Open any BB detail panel > **AI tab** > **Summarise this BB**.

The agent extracts metrics from all uploaded files (S-matrix data, test results, parameters) entirely in-browser, then sends only the computed JSON to the LLM. It never sends raw file bytes unless egress mode is set to full. The resulting summary is cached and reused until you click **Refresh**.

You can also trigger an **Executive summary** for an entire platform or the whole dashboard from the main Insight modal.

9.3 Multi-Turn Chat

Click **Insight > Chat with the agent**.

The agent has access to the following tools, which it calls automatically:

Tool	What it reads
`list_platforms`	All platforms and their BB counts
`list_bbs`	BBs filtered by platform, category, status, or contributor
`get_bb`	Full detail for a specific BB
`list_contributors`	All contributors and their BB counts
`get_bb_metrics`	Computed metrics for a BB
`search_bbs`	Full-text search across names, descriptions, parameters
`summarise_bb`	Generates and caches an AI summary for a BB

Example queries:

- "List every grating coupler on SOI 220 nm with their insertion loss"

- "Which BBs are in progress and have no S-matrix data?"
- "Who contributed the most BBs to the TFLN platform?"
- "Summarise the MMI 1x2 on SOI 340 nm"

Chat history is saved in localStorage and persists across sessions. Use **Clear conversation** to start fresh, or **Export .md** to download the transcript.

9.4 GitHub File Fetcher

Inside the Chat modal, the **Data files in context** panel (click the triangle to expand) lets you load external data files for analysis.

URL type	Example
Single file (blob)	`github.com/org/repo/blob/main/data/results.csv`
Single file (raw)	`raw.githubusercontent.com/org/repo/main/data/results.csv`
Folder	`github.com/org/repo/tree/main/measurements/`
Repo root	`github.com/org/repo`
Multiple URLs	Paste several, one per line or comma-separated

How to use:

11. Paste one or more URLs into the textarea.
12. Click **Fetch** or press Ctrl+Enter / Cmd+Enter.
13. The fetcher resolves each URL, calls the GitHub Contents API for folders (recursive up to 2 levels), and loads every matching data file.
14. Loaded files appear as rows with a checkmark (in context) or empty box (excluded). Click the icon to toggle, or X to remove.

Supported file types fetched from folders: `.csv`, `.tsv`, `.json`, `.yaml`, `.yml`, `.py`, `.md`, `.log`, `.txt`, `.toml`, `.ini`, `.cfg`, `.dat`, `.s2p`, `.s4p`

Tip: For private repos or to avoid the 60 req/hr unauthenticated limit, set your GitHub PAT in the GitHub settings — the fetcher picks it up automatically.

9.5 Analyse Files with AI

Once one or more files are loaded and active, click **Analyse files with AI**.

Step 1 — Client-side pre-analysis (instant, no API tokens used)

- **CSV / TSV:** delimiter detection, column type inference, statistics (count, null%, min, max, mean, median, std, Q1, Q3), IQR outlier detection, Pearson correlations for numeric column pairs
- **JSON:** recursive schema map (4 levels deep), null/empty field detection
- **YAML:** top-level key extraction, null/empty field list

Step 2 — LLM interpretation

The pre-computed report is sent to the AI, which produces:

- A **markdown table** of column statistics per file
- **Anomaly flags**: constant columns, high-null columns (>30%), outlier-rich columns, strong correlations (|r| > 0.5)

- **2-4 insight bullets** focused on what matters for a photonics PDK engineer

The analysis result appears as an agent reply in the chat, so you can follow up naturally.

9.6 Detachable Chat Window

Click **Detach** in the chat footer to lift the chat into a free-floating window.

- **Drag** the title bar to reposition anywhere on screen
- **Resize** from the bottom-right corner handle
- **Minimise** to a small pill at the bottom-right of the screen
- **Dock** to move everything back into the modal
- Window position and size are remembered in localStorage

10. Roles & Access Control

Role	Permissions
Admin	Full access: add/remove/edit all BBs, platforms, users, publish, manage lists
User-admin	Can edit BBs in their own `users/<id>/` folder; read-only for others
Read-only	View all data, run scripts, download files; cannot edit or publish

The role is set in `localStorage` (key: `pdk-monitor.role`). By default the dashboard opens in read-only. To switch to Admin mode, see the comment at the top of the `<script>` block in the HTML file.

11. Import / Export

Export JSON

Downloads the full dashboard model as `dashboard.json`. Use this to back up your data, commit to a GitHub repository, or share with collaborators.

Import JSON

Loads a previously exported `dashboard.json`, replacing the current model. A local backup is saved automatically before replacing.

Export BB folder (zip)

From any BB detail panel > Files tab > **Download folder**: downloads a zip containing all files for that BB, plus an auto-generated `README.md` manifest.

Run locally (zip)

For Python scripts: downloads a self-contained folder with the script, its data files, and a `run.sh` helper, ready to execute from a terminal.

12. Keyboard Shortcuts

Shortcut	Action
/	Focus search box
Escape	Close modal / detail panel
Ctrl+Enter / Cmd+Enter	Send message in AI chat; Fetch files in URL input
Ctrl+S / Cmd+S	(In script editor) Save script

13. Troubleshooting

AI agent returns "No API key"

Open Insight > Settings and enter your API key. Click Test connection to verify.

GitHub file fetch returns 403

You have hit the unauthenticated rate limit (60 req/hr). Add your GitHub PAT in the GitHub settings.

GitHub file fetch returns 404

Check the branch name in your URL. GitHub defaults to `main`; some repos still use `master`.

File fetcher loads 0 files from a folder

The folder may contain no files with supported extensions, or it may be empty. Check the URL points to the correct subfolder.

Storage indicator is red

You are near the localStorage limit (~5-10 MB depending on browser). Export JSON, then use **Clear AI cache** in AI Settings and delete old local backups to free space.

Pyodide script runner times out

Pyodide loads from CDN on first use (~10 MB). Check your network connection. Scripts that import large packages (numpy, scipy) take longer to initialise.

Dashboard shows stale data after a pool fetch

Click the refresh button in the header to reload the model from localStorage.

For questions, issues, or contributions, see the GitHub repositories linked in the dashboard header.